**Summary**

Senior Software Engineer focused on distributed systems, cloud-native software, and Generative AI, with experience delivering scalable microservices and patented solutions at Cisco, and a strong foundation in networking and systems programming

# MARCO TRINELLI



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## Employment

**Senior Software Engineer** — **Cisco Systems (Rolle, CH)** — **Dec. 2019 – now**

- Pioneered and scaled a production-grade GenAI system leveraging AI agents and Retrieval-Augmented Generation (RAG) for real-time enterprise knowledge retrieval amd automated Root Cause Analysis, laying the foundation for a new venture at Cisco – Python, React/TypeScript, LangGraph
- Evaluated and implemented retrieval architectures including Vector DB-based RAG and GraphRAG for complex multi-hop queries; contributed to Neo4j vector search semantic retrieval capabilities
- Deployed and evaluated open-source LLMs using vLLM across multi-host, multi-GPU environments
- Architected and contributed to open-source software in the Linux Foundation enabling framework-agnostic multi-agent runtime and inter-agent communication – Python, Go, LangChain, LlamaIndex, MCP
- Led end-to-end full-stack architecture and development across frontend, backend, and GenAI components, driving execution and unblocking critical technical challenges in collaboration with Product and UX teams
- Designed and deployed cloud-native microservices on AWS, building scalable REST/gRPC APIs, and custom IAM (identity, authentication, RBAC) – Go, React/TypeScript, Docker, Kubernetes, ArgoCD, Jenkins, OPA
- Inventor of 12 patents (6 granted, 6 pending) in distributed systems, cloud computing and Generative AI

**Software Engineer Intern** — **Cisco Systems (San Jose, CA)** — **May 2019 – Nov. 2019**

- Programmed new data-plane features on Cisco Nexus switches with Intel Tofino ASICs, enabling advanced traffic monitoring (via IPinIP and VXLAN encapsulation) and IPv4 to IPv6 transition (MAP-T), significantly increasing network throughput for a major US telecom operator – P4 Language, Python
- Integrated data plane capabilities into control plane (NX-OS) – C++

**R&D Intern, Software Engineer** — **Nokia Bell Labs (Paris, FR)** — **July 2018 – Jan. 2019**

- Developed new components (SOCKS5 proxy, FFmpeg-based video encoder/decoder), allowing legacy hardware to run AR/VR services at the network edge, boosting performance by ~20x FPS – C++
- Publication: Transparent AR Processing Acceleration at the Edge, Marco Trinelli et al., EdgeSys ’19

## Education

**Turin, IT** — **Polytechnic University of Turin** — **Sept. 2016 – Apr. 2019**

- M.Sc. Computer Engineering – Computer Networks
- Focus: Computer Networks and Data Centers, Cloud Computing, System Programming
- Grade: Summa cum laude

**Biot, FR** — **EURECOM - Telecom ParisTech** — **Sept. 2017 – Jan. 2019**

- M.Sc. Computer Engineering – Communication System Security
- Focus: Network Security, Distributed Systems, Cloud Computing, Software Defined Networking (OpenFlow)

## Projects and Patents

- **Notification Service – US Patent 12,368,768** (2025). Designed a distributed microservice adopted across Cisco, scaling web applications with real-time updates from stateless backend services (REST, gRPC, …), and improving system throughput by ~10x in multi-tenant environments – Go
- **OS161 – Unix-like OS** (2017). Implemented system calls, file system, virtual memory management, and locks – C

## Languages and Skills

- Languages: Italian (native), English (fluent), French (fluent), Portuguese (fluent)
- Tech: Python, Go, C/C++, React, TypeScript, Kubernetes, Docker, LLMs, Networking Protocols, vLLM, MCP
- Soft skills: Effective Communication, Leadership, Teamwork & Collaboration